

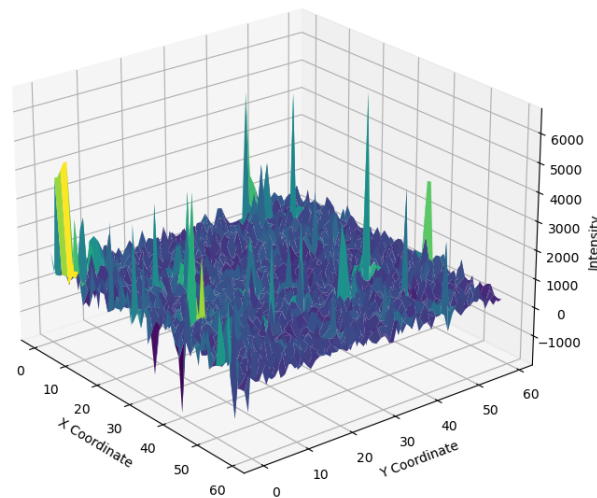
Overview:

Raman spectroscopy is a technique used to analyze molecular structures, this project aims to understand the composition of Raman spectra data of brain tissue and classify its cancerous regions using a K-means clustering algorithm. Raman spectra data allows us to detect subtle molecular differences of the tissue while maintaining the spatial resolution within the tissue sample. K-means, an unsupervised learning method, is used as the first level of classification of the data as it provides rapid analysis and can effectively reduce the high dimensionality of the data while still capturing underlying patterns.

Data Processing:

Initially, wavelengths 0-104 and 907-1374 were cut from the dataset as is standard process for the type of machine used. The following is the visual representation of raw data at a particular spectral point:

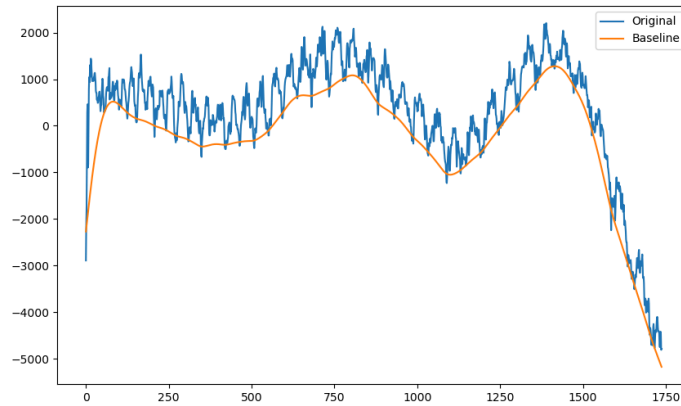
3D Surface Plot of Intensity at Spectral Point 687 (Raw Data)



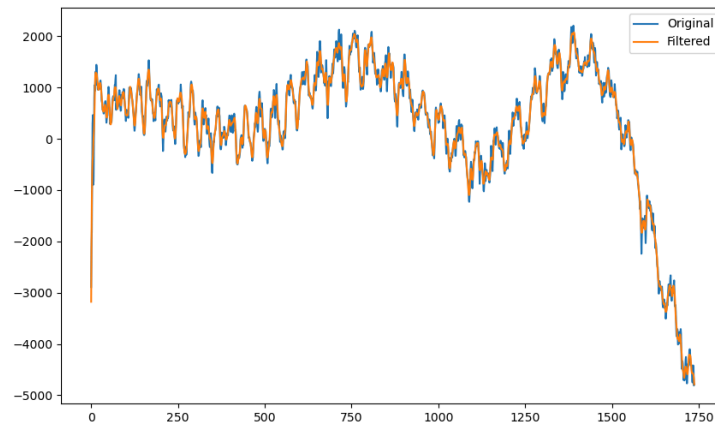
To further process the data, it was necessary to set all negative values to 0 as it is unclear why they appear in the data set. The following steps were applied:

- Savitzky-Golay filter: to smooth the data and reduce noise while preserving peak shapes
- Baseline subtraction: to remove background interference and improve peak visibility
- SVN normalization: to standardize the intensity scale across all spectra, allowing for fair comparison"

Baseline subtraction is done to reduce noise and provide a 'baseline' for the data, as seen below.

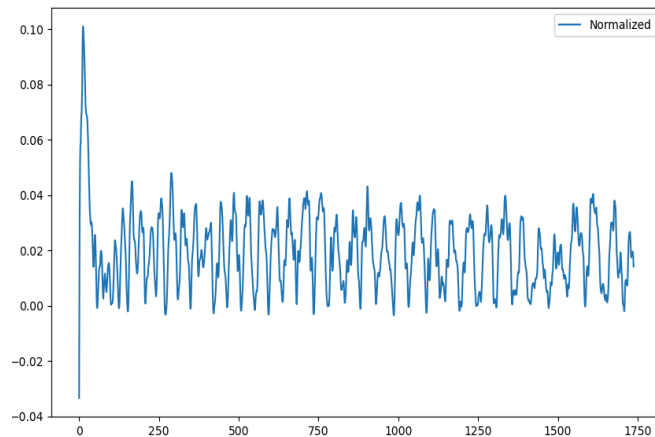
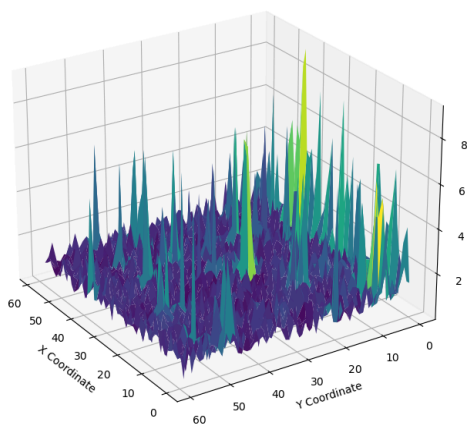


Furthermore, as seen below for a particular spectral point, Noise reduction was done to minimize noise in the K-means analysis.



The final step is normalization of the data. Normalization is necessary for K-means analysis and will not negatively impact results as we are interested in which wavelengths have peaks rather than the intensity of each peak. Below is a 3D and 2D sample of a fully processed spectral point.

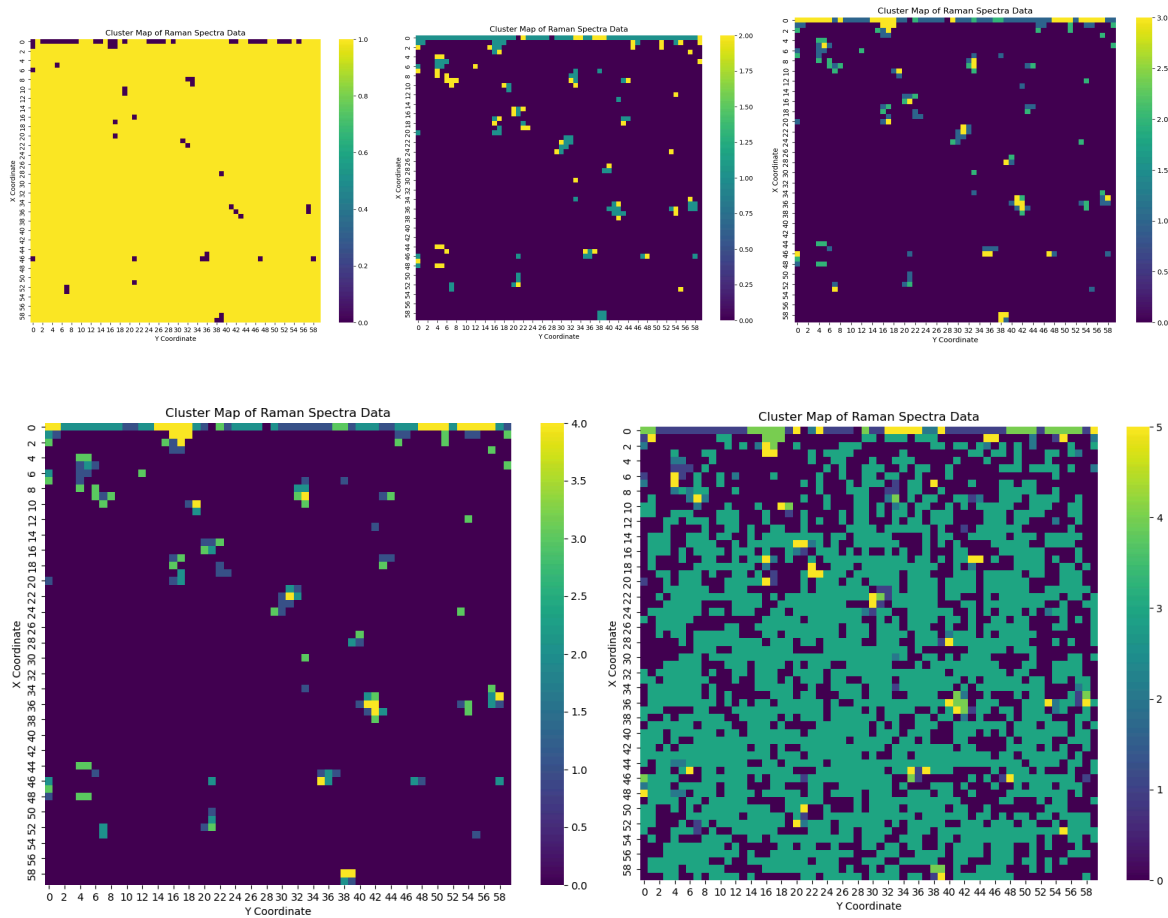
3D Surface Plot of Intensity at Spectral Point 583 (Preprocessed Data)



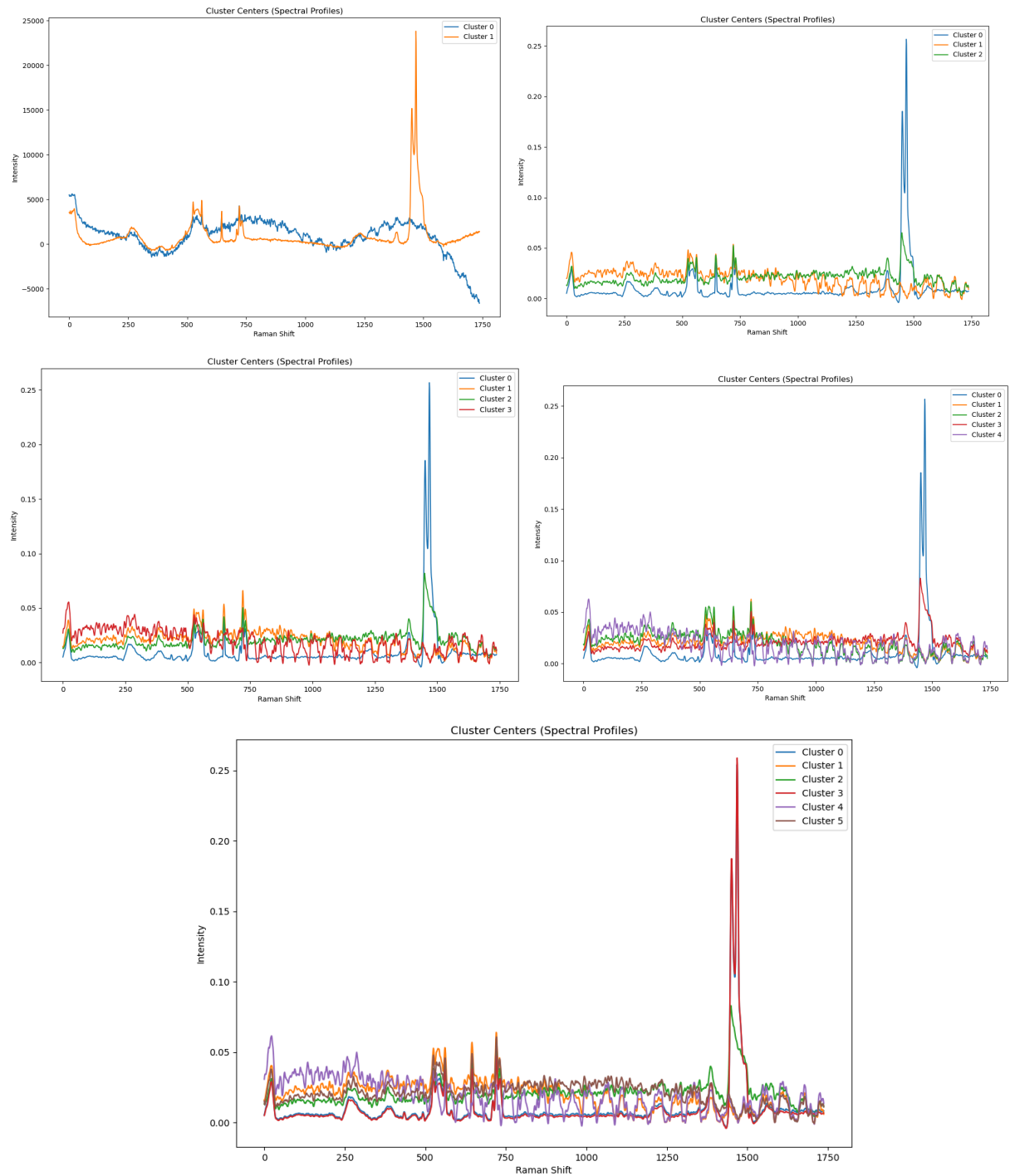
Modeling:

As described, a K-means model clustering algorithm is used for this project as it allows for a characterization for each data point as healthy or a type of tumor leveraging the algorithm's ability to group similar spectra together. This can help identify distinct spectral patterns associated with healthy tissue and different tumor types, potentially revealing subtle molecular differences that distinguish these tissue states.

Thus, a K-means clustering algorithm was run for 2, 3, 4, 5, and 6 clusters at a time. The following are the results of each cluster size respectively.



The following are spectral profiles of each clusters in each of the previous maps:



Model Evaluation:

When examining the cluster maps, we observed that the representations for cluster sizes 3 through 5 were remarkably similar, showing little variation in the spatial distribution of cluster

assignments. This consistency suggests that increasing the number of clusters beyond 3 did not provide significant additional discriminatory power in segmenting our data.

Notably, when we applied 6 clusters to our dataset, the resulting clusters did not appear to correspond to any meaningful patterns in the underlying spectral data. This observation indicates a potential overfitting of the model at higher cluster numbers, where the algorithm may be fitting to noise rather than genuine spectral features that distinguish between tissue types.

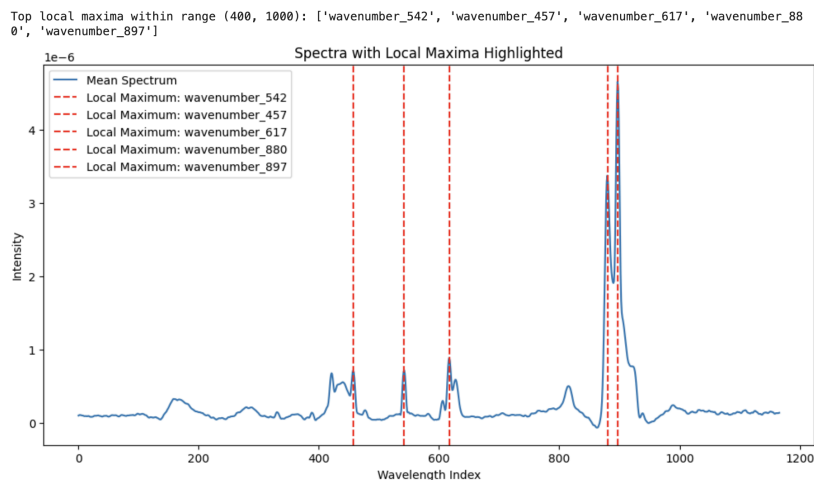
To further validate our K-means clustering results and quantitatively assess the optimal number of clusters, we performed silhouette analysis on our Raman spectra dataset which provides a measure of how well each data point fits within its assigned cluster compared to other clusters.

We calculated the average silhouette score for each clustering configuration (2 to 6 clusters) to evaluate the overall quality of the clustering. This analysis showed that 4 clusters allowed for the optimal balance of cluster amount and variance within each cluster. Additionally, when compared to results received from work of a previous group, the cluster locations closely match those of this analysis. However, the next steps to improve clustering algorithms on this dataset is to further attempt to remove noise from the dataset- a common downside of raman spectroscopy.

Further Analysis:

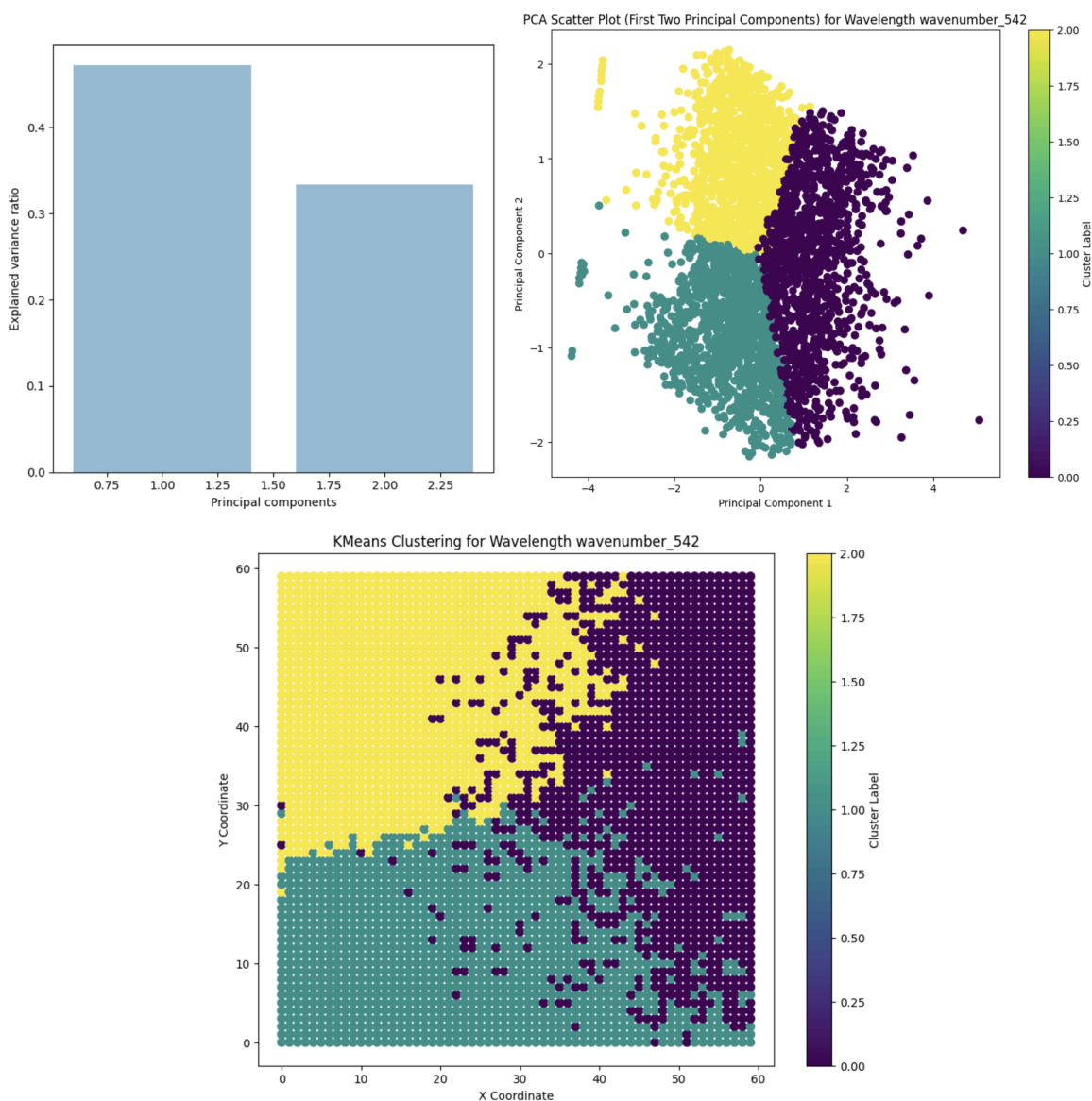
As mentioned, to further attempt to remove noise from the data without erasing potential patterns, an algorithm was created to determine particular wavelengths within notable wavelength sections of the data to determine which areas provide meaningful input for the K-means.

The first step of this process is determining which areas should be investigated as candidates for this search. To do so, an algorithm to search for local maxima within the dataset (for a particular spectral point) was created. This process identifies peaks of the dataset which correspond to unique chemical markers.

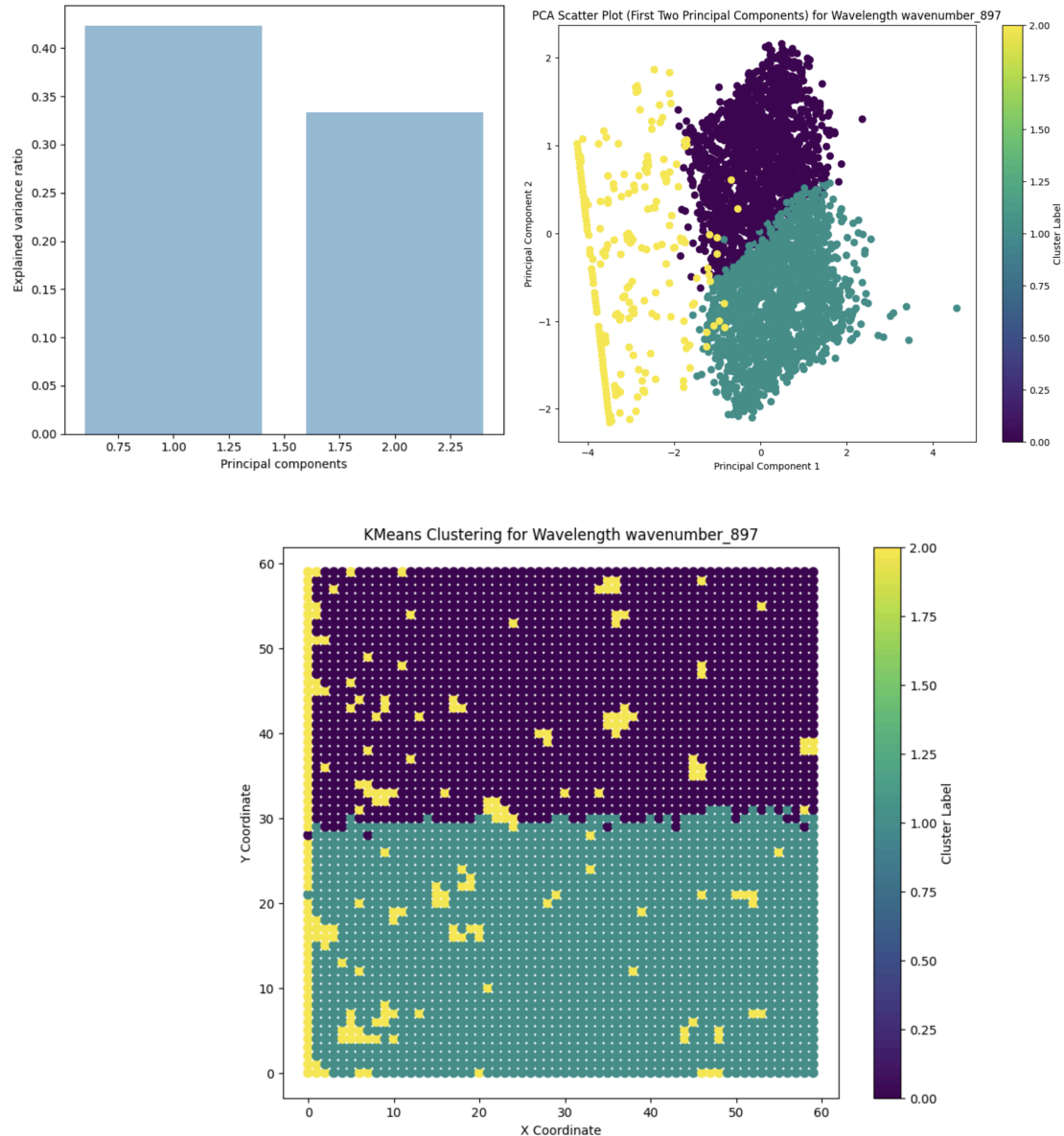


Then, a smaller K-means was run on the data for each of the identified wavelengths. The K-means implemented a silhouette score loop that would output the K-means which obtained the score for the clusters, and a PCA was run on the data prior to modeling. These steps were taken to attempt to characterize the wavelengths which are deemed important for analysis so that future modeling can only consider wavelengths that receive similar K-means or PCA results.

From these steps, 2 main types of wavelengths were identified. The first type showed no conclusive pattern from the K-means (besides simply forming clusters based on the spatial location of the data) and the principal components of the PCA did not account for a meaningful amount of variance. Furthermore, the silhouette score of such wavelengths remained around .200 to .300, which shows that each cluster does not capture much unique information.



The second type of wavelength seen appears to much more closely match the results of K-means when run on the entire dataset. Additionally, the silhouette scores of these wavelengths typically remained above .300. However, the PCA results remained inconclusive. The results of the analyses performed is seen below:



These are promising results as they give us insight into the analytic understanding of which wavelengths of the dataset contain the markers of tumors and, thus, which wavelengths should be used for clustering algorithms run on raman spectra data.

Conclusions:

Based on our analysis of Raman spectra data from brain tissue samples using K-means clustering and additional analytical techniques, we can draw the following conclusions:

1. K-means clustering effectively grouped similar spectra, potentially distinguishing between healthy and tumorous tissues. The optimal number of clusters appears to be between 3 and 4, balancing meaningful distinctions and overfitting risk.
2. Our wavelength selection algorithm identified specific regions of the Raman spectrum that contain more meaningful information for tumor detection. This insight can guide future analyses by focusing on these key spectral regions.
3. The combination of K-means clustering, silhouette analysis, and targeted wavelength investigation provides a promising approach for analyzing Raman spectra of brain tissues and potentially identifying cancerous regions.
4. While our method shows promise, further refinement is needed to address the inherent noise in Raman spectroscopy data and improve the accuracy of tumor detection.

Future work should focus on:

- Validating our findings with histopathological data
- Developing more sophisticated noise reduction techniques
- Exploring advanced machine learning algorithms that can leverage the identified key spectral regions
- Investigating the biological significance of the spectral patterns associated with each cluster

This study demonstrates the potential of Raman spectroscopy combined with machine learning techniques for non-invasive brain tumor detection and classification. With further development, this approach could become a valuable tool in clinical diagnostics and surgical guidance.